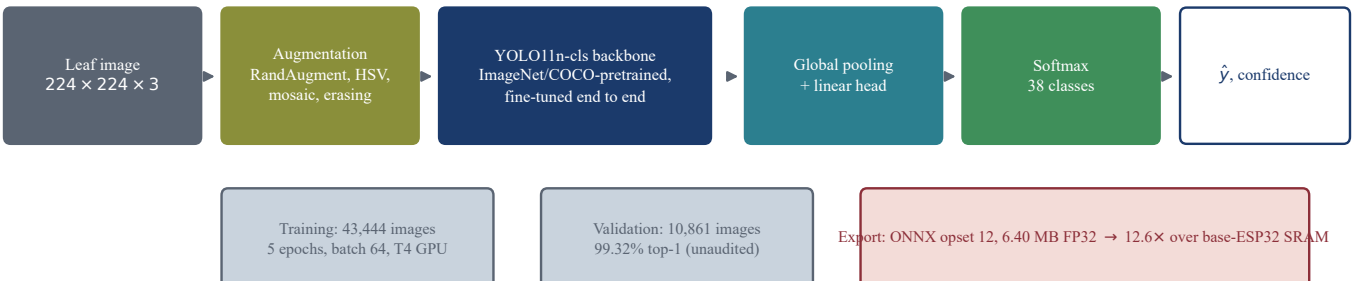
Leaf image
224 × 224 × 3
Augmentation
RandAugment, HSV,
mosaic, erasing
YOLO11n-cls backbone
ImageNet/COCO-pretrained,
fine-tuned end to end
Global pooling
+ linear head
Softmax
38 classes
ŷ, confidence
Training: 43,444 images
5 epochs, batch 64, T4 GPU
Validation: 10,861 images
99.32% top-1 (unaudited)
Export: ONNX opset 12, 6.40 MB FP32 → 12.6× over base-ESP32 SRAM